

CAPTURE	
/astro status	State, camera, active profile
/astro capture	One frame at profile settings
/astro capture frames=N exposure=S iso=N	Capture with overrides
/astro preview exposure=S iso=N	Quick single frame
/astro last	Resend last preview
/astro abort	Stop current session
Abort waits. It stops between frames — a 300 s exposure already running will finish first.	

TIMELAPSE	
/astro timelapse profile=<name>	Run a named profile
/astro timelapse profile=<name> duration=S	Profile, any value overridden
/astro timelapse interval=S frames=N	Manual, by frame count
/astro timelapse interval=S duration=S	Manual, by total time

PROFILE	RUN	EVERY	ISO	EXPOSURE
sunset	90 min	6 s	100→3200	0.5→4 s
sunrise	90 min	6 s	3200→100	4→0.5 s
blue-hour	40 min	6 s	400→1600	1→3 s
night-sky	2 hr	30 s	3200	15 s
stars	3 hr	30 s	6400	20 s
golden-hour	60 min	5 s	100	0.01 s
Twilight profiles ramp ISO and exposure logarithmically, then hold at the end values — sunset ramps 50 min, holds 40.				

EXPOSURE RAMPING (MANUAL)	
iso_start= iso_end=	ISO at first and last ramp frame
exposure_start= exposure_end=	Exposure across the ramp
ramp_duration=S	Ramp seconds, then hold

TIMELAPSE PROFILES	
/astro profiles	List built-in + custom
/astro profile add <name> k=v...	Save a custom profile
/astro profile rm <name>	Delete a custom profile

CAMERA & CAPTURE PROFILE	
/astro camera	Camera, limits, defaults
/astro camera a6400 a6700	Switch camera
/astro profile	Show active profile
/astro profile frames=N exposure=S iso=N	Update active profile
/astro defaults	Show camera defaults
/astro defaults k=v	Save new defaults
/astro defaults load	Reset profile to defaults
/astro battery	Battery level
/astro limits	Exposure and ISO limits
/astro exposures	Valid shutter speeds
/astro isos	Valid ISO values

PRESETS (CAPTURE PROFILE SNAPSHOTS)	
/astro preset list	List saved presets
/astro preset save <name>	Save current profile
/astro preset load <name>	Load into active profile
/astro preset show <name>	Show settings
/astro preset delete <name>	Delete preset

SESSIONS, ARCHIVE, RENDER	
/astro sessions	Sessions on the Pi
/astro shots <YYYYMMDD>	Frames + thumbnails
/astro archive session <YYYYMMDD>	Archive session to LXD
/astro archive last <id>	Archive one frame
/astro archived [session]	Browse the LXD archive
/astro render <YYYYMMDD_HHMMSS>	Stitch a run into MP4
/astro delete session <YYYYMMDD> confirm	Delete a session
/astro delete all confirm	Delete all shots

BEFORE A SESSION	
- Camera on, dial to M (Manual). - Menu → Setup → USB Connection → PC Remote. If it shows as “charging mode”, gphoto2 cannot see it. /astro battery to confirm the link.	

RULES OF THUMB	
500 rule. Max exposure before star trails $\approx 500 \div$ focal length. A 20 mm lens → ~25 s. Bulb is automatic. Over 30 s switches to bulb with no action needed. Mount control is not implemented — /astro track replies “Mount not connected yet”.	